SIMATS ENGINEERING
ASSESSMENT TOOL 3: Mock Interview

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED
CO5: *Design big data processing systems using the Hadoop ecosystem including HDFS, MapReduce, and YARN.* (BL5)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 25**
Question Count: 10

Code of Conduct
I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: *G.Hima Bindu*

Assessment Tasks
Mock Interview
Answer the following interview-oriented questions clearly and concisely. Support your responses with architecture-level reasoning wherever appropriate.

1. Explain the difference between HDFS and traditional file systems.
2. Why is replication important in distributed storage systems?
3. How does MapReduce divide work between map and reduce phases?
4. What role does YARN play in large-scale Hadoop processing?
5. Differentiate NAS and SAN in an enterprise data environment.
6. What is the purpose of ETL in a big data pipeline?
7. How does Apache NiFi help in data integration workflows?
8. What are the key failure points in a Hadoop cluster and how would you handle them?
9. How would you design a secure data ingestion pipeline for a large organization?
10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Mock Interview Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Knowledge	25%	Shows deep and accurate technical understanding	Good technical understanding	Basic understanding	Weak understanding
Problem Solving	20%	Responds with strong architecture-level reasoning	Reasonable reasoning	Basic reasoning	Weak reasoning
Communication	20%	Answers are confident, precise, and professional	Mostly clear communication	Moderate clarity	Weak communication
Practical Awareness	20%	Connects concepts to real-world implementation well	Some practical relevance	Limited practical relevance	No practical linkage
Confidence and Professionalism	15%	Highly confident and professional	Generally professional	Moderately professional	Low confidence and professionalism

1. Explain the difference between HDFS and traditional file systems.

Answer:
HDFS stands for Hadoop Distributed File System. The main difference is that a traditional file system generally stores data on a single computer or centralized storage, whereas HDFS stores data across multiple computers in a Hadoop cluster.

HDFS divides large files into blocks and distributes those blocks among different Data Nodes. It also keeps multiple copies of the blocks, so even if one machine fails, the data is still available.

Traditional file systems are mainly designed for normal applications, while HDFS is specifically designed for very large datasets and distributed processing.

So, in simple terms, I would say HDFS provides distributed storage, fault tolerance and scalability, whereas a traditional file system is mainly designed for individual or centralized storage.

2. Why is replication important in distributed storage systems?

Answer:
Replication means keeping multiple copies of the same data on different machines.

It is important because failures are common in a large distributed system. For example, if one DataNode fails, the data stored on that node should not become unavailable.

In HDFS, multiple copies of a data block are maintained. If one copy is lost, another copy can be used. Replication therefore improves data availability, reliability and fault tolerance.

It can also improve performance because data can sometimes be read from a nearby replica.

So, I would say replication is mainly used to protect data from hardware failures and keep the system running continuously.

3. How does MapReduce divide work between map and reduce phases?

Answer:
MapReduce divides a large data processing task into smaller tasks so that they can be processed in parallel.

In the Map phase, the input data is divided into smaller portions, and each Mapper processes its portion and produces key-value pairs.

Then comes the Shuffle and Sort phase, where similar keys are grouped together.

Finally, in the Reduce phase, the Reducer processes these grouped values and generates the final result.

For example, in a word-count problem, the Mapper identifies each word and gives it a count of one. The Reducer then adds the counts for each word.

So, simply, Map transforms the input data and Reduce combines or summarizes the results.

4. What role does YARN play in large-scale Hadoop processing?

Answer:
YARN stands for Yet Another Resource Negotiator. Its main job is to manage the resources of a Hadoop cluster.

It decides how much CPU and memory should be given to different applications and where their tasks should run.

The main components include the Resource Manager, Node Manager and Application Master.

For example, if multiple Hadoop jobs are running at the same time, YARN manages the available resources and schedules those jobs properly.

So, I would explain YARN as the resource manager or coordinator of a Hadoop cluster. It helps improve resource utilization and allows multiple applications to run on the same cluster.

5. Differentiate NAS and SAN in an enterprise data environment.

Answer:
NAS stands for Network Attached Storage, while SAN stands for Storage Area Network.

NAS provides file-level storage. Users or applications access files and folders over a network. It is commonly used for file sharing, backups and document storage.

SAN provides block-level storage. To the server, the storage appears more like a local disk. SAN is generally used for high-performance applications such as databases and enterprise systems.

So, the simple difference is: NAS is mainly for file sharing, while SAN is mainly for high-performance block storage.

NAS is generally simpler to manage, while SAN can provide higher performance but is more complex.

6. What is the purpose of ETL in a big data pipeline?

Answer:
ETL stands for Extract, Transform and Load.

First, Extract means collecting data from different sources such as databases, applications, files or APIs.

Second, Transform means cleaning and modifying the data. For example, we can remove duplicate records, handle missing values and convert data into a common format.

Finally, Load means storing the processed data in a target system such as a data warehouse, data lake or Hadoop storage.

The main purpose of ETL is to convert raw and unorganized data into clean and useful data for analysis.

So, ETL helps an organization prepare data before using it for reporting, analytics or decision-making.

7. How does Apache NiFi help in data integration workflows?

Answer:
Apache Ni Fi is a tool used for data integration and data flow management.

It can collect data from different sources, process it and send it to different destinations. For example, it can collect data from a database, transform it and send it to Hadoop or cloud storage.

One important feature of Ni Fi is its visual interface. We can create data flows using processors and connections without writing a large amount of code.

It also provides monitoring, data routing, transformation and data provenance.

So, in an interview, I would say Apache Ni Fi makes it easier to move, transform, monitor and manage data between different systems.

8. What are the key failure points in a Hadoop cluster and how would you handle them?

Answer:
There can be several failure points in a Hadoop cluster, such as DataNode failure, disk failure, NameNode failure, network failure and job failure.

For DataNode or disk failures, HDFS replication helps because multiple copies of the data are available.

For NameNode failure, we can use High Availability with active and standby NameNodes.

For job failures, Hadoop can retry failed tasks. Network problems can be handled using proper monitoring and redundant network connections.

I would also continuously monitor the cluster for CPU, memory, disk and network problems.

So, the main approach is replication, high availability, monitoring, automatic recovery and proper backups.

9. How would you design a secure data ingestion pipeline for a large organization?

Answer:
First, I would make sure that only authorized users and applications can send data into the system.

I would use authentication and authorization mechanisms to control access. During data transfer, I would use encryption such as TLS to protect the data.

After receiving the data, I would validate it to check for incorrect, duplicate or suspicious records. Sensitive data should also be encrypted when stored.

I would use tools such as Apache NiFi for controlled data movement and monitoring. I would also maintain audit logs so that we can track who accessed or modified the data.

So, the important security measures are authentication, authorization, encryption, data validation, monitoring, auditing and secure storage.

10. How would you explain a scalable Hadoop-based processing system to a non-technical manager?

Answer:
I would explain Hadoop as a team of computers working together to handle a very large amount of data.

For example, suppose a company has millions of customer transactions. Instead of asking one computer to process everything, Hadoop divides the data and distributes the work among many computers.

HDFS stores the data across these computers and keeps copies to protect against failures. YARN manages the available resources, while MapReduce or another processing framework performs the actual data processing.

The important point is scalability. If the company gets more data, we can add more machines to the cluster instead of replacing the existing system with one very powerful machine.

So, in simple words, Hadoop allows a company to store and process growing amounts of data by making many computers work together efficiently and reliably.